

GARDEN RIVER, AB, Canada

WMO: 712530

Lat: 58.7108N

Lon: 113.8664W

Elev: 790

StdP: 14.28

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-39.6	-34.1	-47.4	0.4	-40.1	-41.7	0.5	-34.9	11.3	12.8	9.9	10.1	0.4	230	0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.7	84.4	66.2	81.1	64.4	77.9	63.0	68.9	80.0	67.1	77.3	65.1	74.1	4.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.2	95.9	73.2	63.3	89.5	71.1	61.4	83.8	69.3	33.6	80.6	32.0	77.5	30.4	74.3	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.4	9.8	8.6	-43.3	90.5	4.6	2.8	-46.6	92.5	-49.3	94.2	-51.9	95.8	-55.2	97.8
			-44.8	72.3	3.7	3.2	-47.4	74.5	-49.5	76.4	-51.6	78.1	-54.3	80.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	31.4	-4.8	2.7	15.3	35.2	48.9	59.0	63.3	59.0	49.1	34.4	13.9	-0.8
	DBStd	26.42	15.82	13.39	15.48	10.77	8.71	6.26	5.23	6.12	7.26	8.01	12.03	15.00
	HDD50	7932	1699	1325	1074	450	124	8	0	5	102	487	1084	1574
	HDD65	12333	2164	1745	1539	895	502	193	94	199	479	950	1534	2039
	CDD50	1147	0	0	0	5	88	279	413	286	74	2	0	0
	CDD65	73	0	0	0	0	1	14	42	15	1	0	0	0
	CDH74	1672	0	0	0	5	150	407	696	366	47	1	0	0
	CDH80	406	0	0	0	0	31	89	190	88	8	0	0	0
	WSAvg	3.2	2.3	2.6	3.6	4.5	4.4	3.5	3.0	3.0	3.0	3.3	2.7	2.3
	PrecAvg	16.4	0.7	0.6	0.7	0.7	1.4	2.5	3.1	2.2	1.7	1.1	0.9	0.8
Precipitation	PrecMax	21.6	1.4	1.4	1.2	1.3	2.2	6.1	5.2	3.7	4.0	1.9	1.5	1.4
	PrecMin	13.6	0.2	0.2	0.2	0.3	0.4	0.7	2.0	1.2	0.6	0.2	0.5	0.3
	PrecStd	2.0	0.3	0.2	0.3	0.2	0.4	0.9	0.8	0.6	0.7	0.4	0.3	0.3
	PrecStd	2.0	0.3	0.2	0.3	0.2	0.4	0.9	0.8	0.6	0.7	0.4	0.3	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.2	41.0	55.8	71.1	83.6	87.1	89.7	87.9	79.8	63.3	44.0	37.2
		MCWB	34.3	34.7	42.3	51.7	59.8	65.3	67.8	69.8	64.1	49.8	38.0	35.0
	2%	DB	29.5	33.3	49.1	63.9	78.1	82.4	84.7	82.1	73.1	56.5	37.4	29.9
		MCWB	28.3	29.9	38.8	47.5	56.5	63.6	67.1	67.2	60.3	46.5	33.8	28.6
	5%	DB	23.3	27.8	43.8	58.8	73.0	78.8	81.5	77.9	68.1	51.3	33.1	24.1
		MCWB	22.6	25.4	35.9	44.9	54.5	61.6	65.9	65.0	57.2	44.2	31.8	23.6
	10%	DB	17.1	22.3	37.9	54.4	67.8	75.1	77.7	74.2	63.5	46.7	30.1	19.6
		MCWB	16.4	21.0	32.3	42.6	51.3	60.5	64.8	63.0	54.7	41.8	29.3	19.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.0	34.8	42.8	51.9	62.4	68.8	72.0	71.3	65.5	52.7	38.1	35.5
		MCDB	36.4	40.1	55.1	67.7	77.9	81.6	82.3	82.5	77.9	61.0	42.4	37.7
	2%	WB	27.8	29.8	39.3	48.3	58.2	66.1	69.5	68.4	61.7	47.4	33.7	29.2
		MCDB	28.9	32.8	48.1	61.3	73.1	76.7	80.6	78.9	70.7	54.8	35.5	30.6
	5%	WB	21.3	25.4	36.1	45.5	54.9	63.9	67.7	66.1	58.6	44.9	31.5	24.3
		MCDB	22.0	27.5	43.1	56.9	68.9	73.7	78.2	74.9	65.7	51.1	32.7	25.0
	10%	WB	14.7	21.1	32.6	42.9	52.3	61.7	66.0	63.6	56.0	42.3	28.7	19.1
		MCDB	15.2	22.6	37.9	53.3	65.0	71.6	75.3	71.4	62.0	46.9	29.6	19.6
Mean Daily Temperature Range	5% DB	MDBR	18.3	24.4	28.1	26.4	27.6	27.2	25.7	25.9	24.4	17.8	15.5	16.9
		MCDBR	23.0	30.0	32.2	34.5	38.3	34.8	32.9	32.7	33.2	27.1	17.2	20.0
		MCWBR	22.9	25.7	22.2	19.8	18.1	16.8	16.7	18.3	19.3	17.7	14.4	19.6
	5% WB	MCDWR	23.8	29.4	30.9	32.4	32.8	29.0	29.7	30.0	29.8	26.1	16.1	21.0
		MCWBR	22.2	25.3	21.6	19.2	16.3	15.6	16.7	18.3	18.8	17.2	13.5	19.6
		MCWBR	22.2	25.3	21.6	19.2	16.3	15.6	16.7	18.3	18.8	17.2	13.5	19.6
Clear-Sky Solar Irradiance	taub		0.217	0.242	0.270	0.329	0.343	0.368	0.387	0.350	0.316	0.286	0.232	0.197
	taud		2.224	2.319	2.306	2.336	2.428	2.376	2.364	2.463	2.542	2.558	2.336	2.117
	Ebn at Noon		207	260	281	279	282	275	266	270	264	242	202	171
	Edh at Noon		13	21	28	33	33	35	35	29	23	17	12	10
All-Sky Solar Radiation	RadAvg		173	473	1008	1547	1810	1869	1747	1424	945	461	183	100
	RadStd		13	25	53	90	148	136	88	75	70	54	21	9

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page